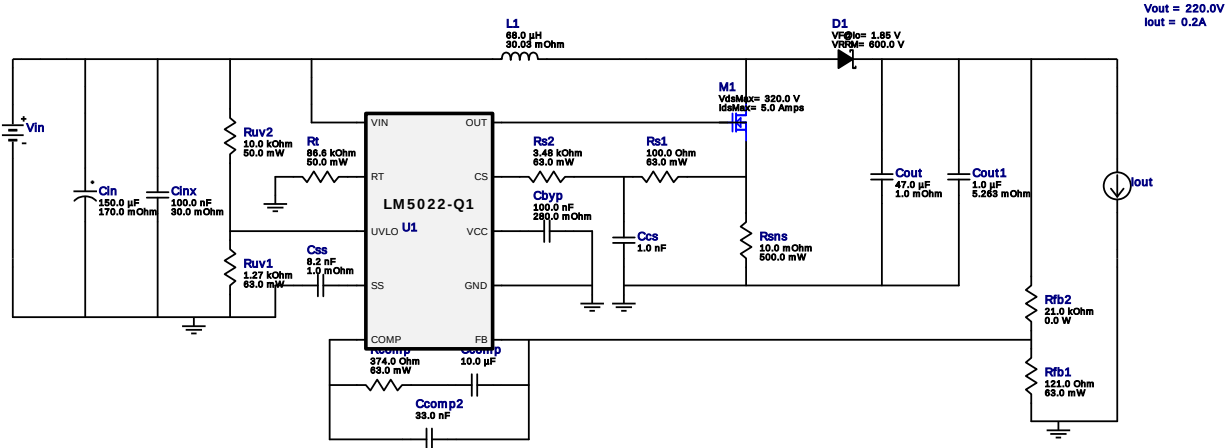


WEBENCH® Design Report

Design : 12 LM5022QDGSTQ1
LM5022QDGSTQ1 14V-17V to 220.00V @ 0.2A



1. This regulator device is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application. View WEBENCH(R) Disclaimer.

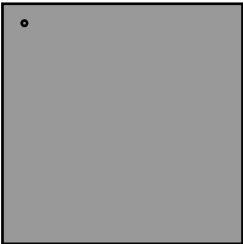
Design Alerts

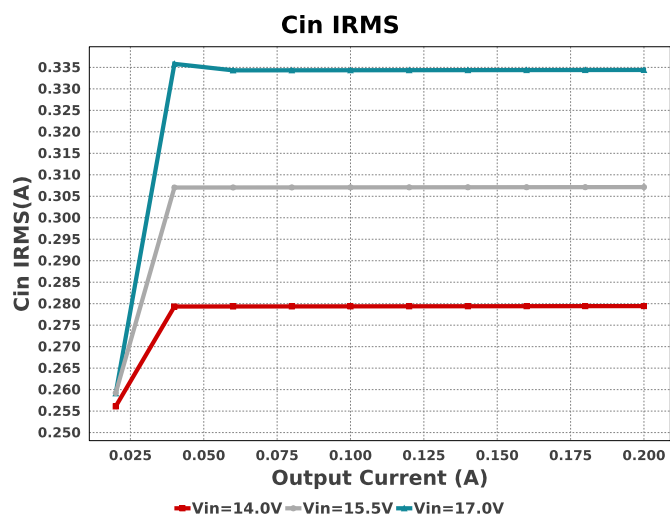
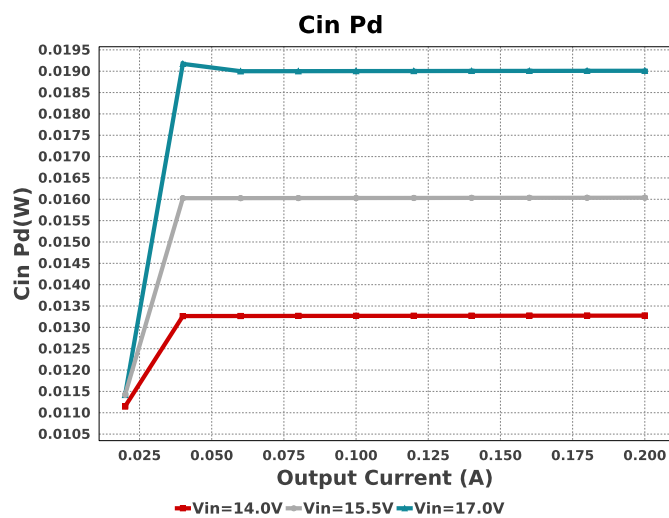
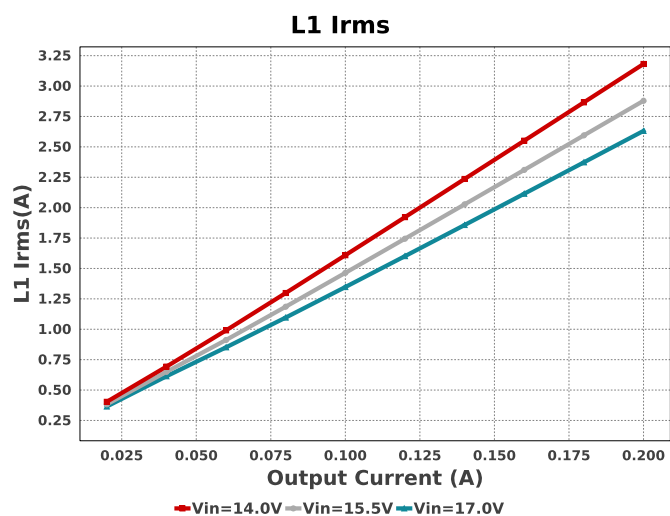
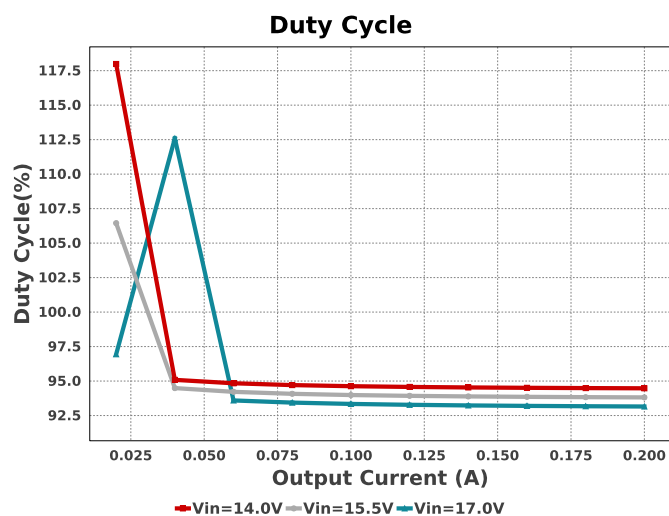
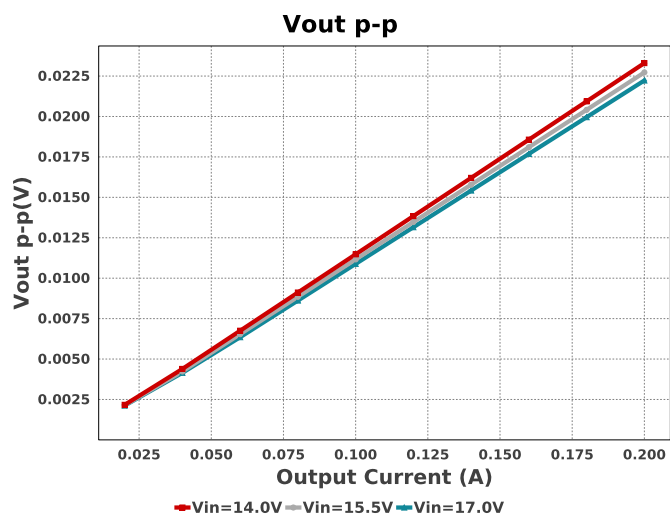
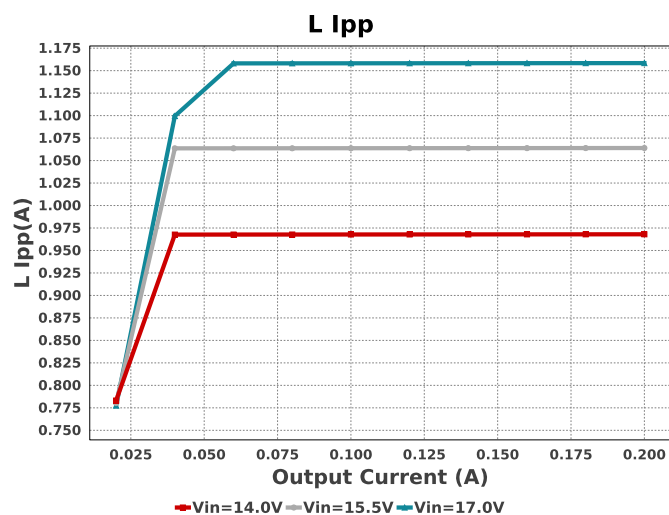
Component Selection Information

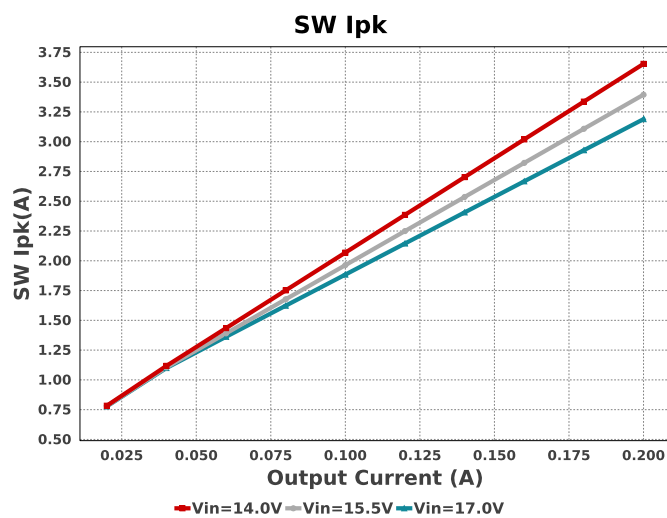
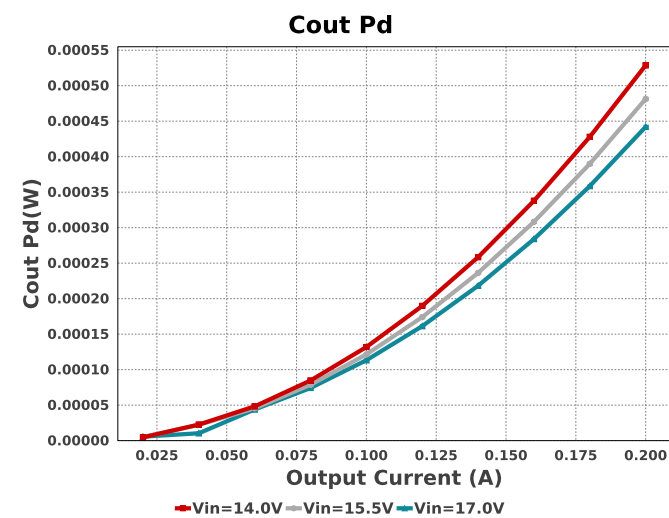
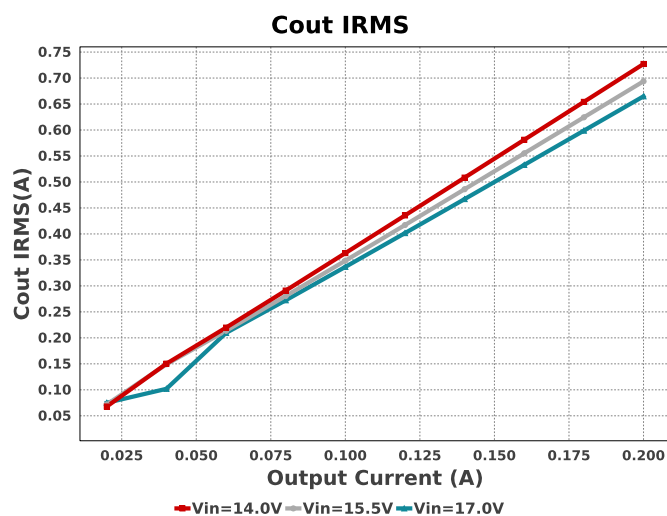
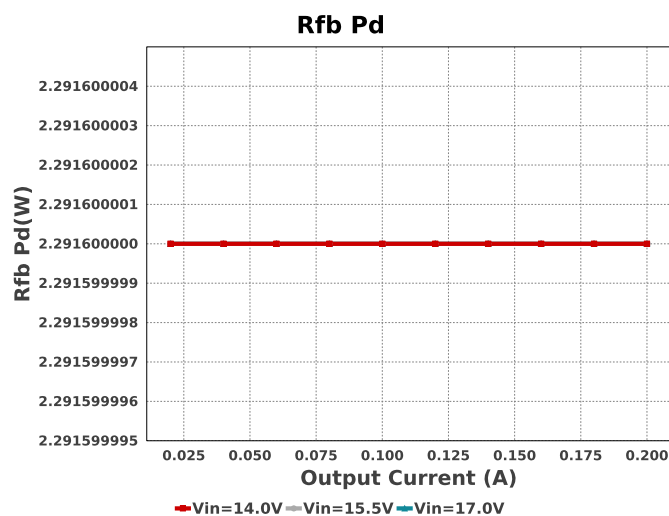
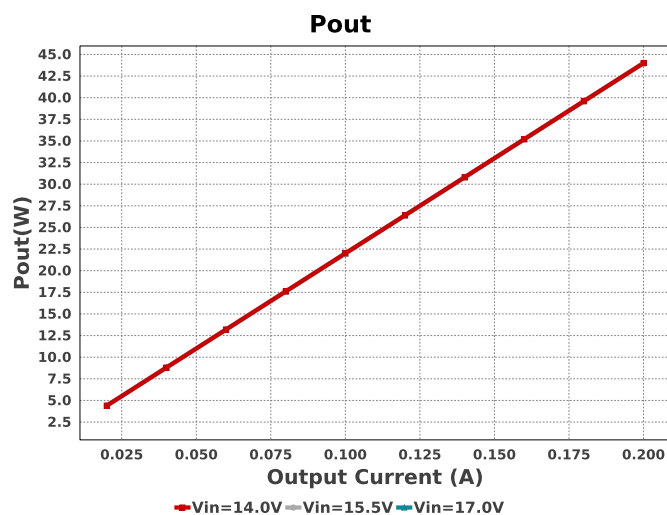
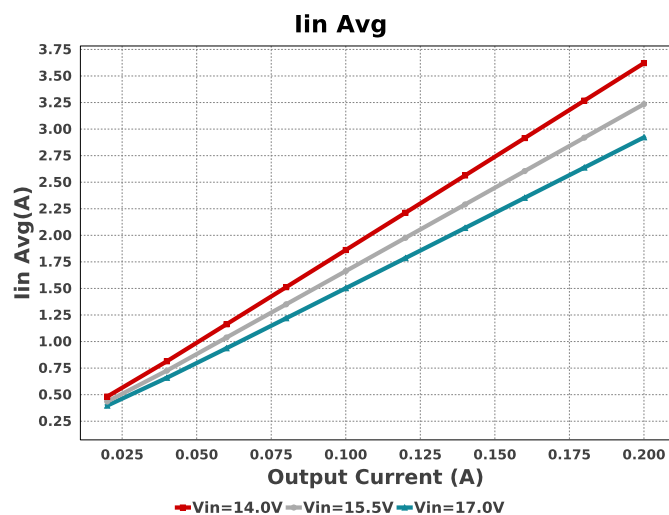
The LM5022-Q1 is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application. Also, with the design conditions, either the IC or the selected FET junction temperature is exceeded above the maximum rating. Hence, this design is created using an ideal FET. Please note that the resulting FET parameters are ideal, so the efficiency/loss opvals have been disabled. Also, the schematic/PCB export and Thermal simulations will not work with the ideal FET.

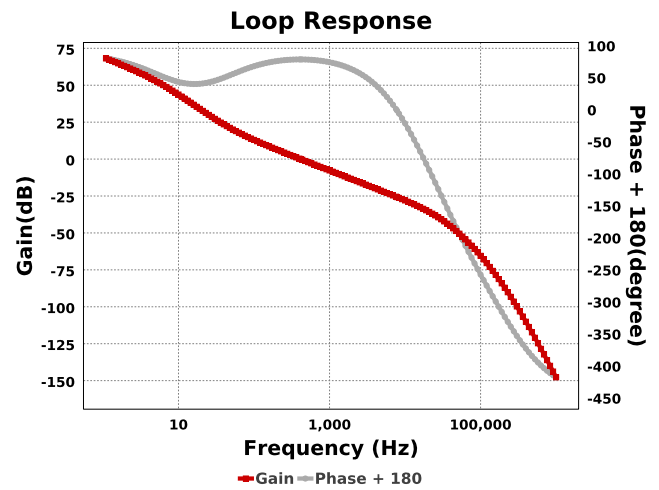
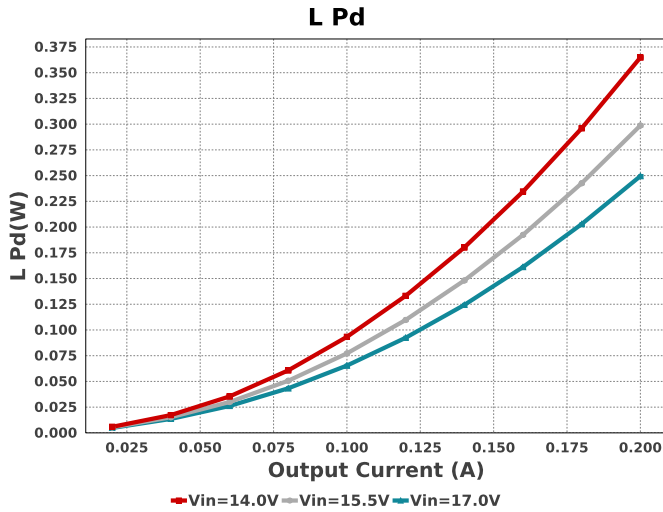
Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbyp	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	\$0.01	0805 7 mm ²
Ccomp	Samsung Electro-Mechanics	CL10A106MQ8NNNC Series= X5R	Cap= 10.0 uF VDC= 6.3 V IRMS= 0.0 A	1	\$0.02	0603 5 mm ²
Ccomp2	TDK	CGA4J2C0G1H333J125AA Series= C0G/NP0	Cap= 33.0 nF VDC= 50.0 V IRMS= 0.0 A	1	\$0.08	0805 7 mm ²
Ccs	Samsung Electro-Mechanics	CL21C102JBCNNNC Series= C0G/NP0	Cap= 1.0 nF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	0805 7 mm ²
Cin	Nichicon	UUD1V151MNL1GS Series= uD	Cap= 150.0 uF ESR= 170.0 mOhm VDC= 35.0 V IRMS= 450.0 mA	1	\$0.23	SM_RADIAL_8MM 113 mm ²
Cinx	MuRata	GRM188R61E104KA01D Series= X5R	Cap= 100.0 nF ESR= 30.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	\$0.01	0603 5 mm ²
Cout	AVX	FFWE6N0356K7X Series= 2237	Cap= 47.0 uF ESR= 1.0 mOhm VDC= 900.0 V IRMS= 67.0 A	1	\$67.61	FFWE_8450X6400 0 mm ²

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cout1	TDK	C5750X6S2W105K Series= X6S	Cap= 1.0 uF ESR= 5.263 mOhm VDC= 400.0 V IRMS= 0.0 A	1	\$1.25	 2220 54 mm ²
Css	MuRata	GRM155R71E822KA01D Series= X7R	Cap= 8.2 nF ESR= 1.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	\$0.01	 0402 3 mm ²
D1	STMicroelectronics	STTH506B-TR	VF@Io= 1.85 V VRRM= 600.0 V	1	\$0.99	 DPAK 102 mm ²
L1	Würth Elektronik	74435586800	L= 68.0 µH 30.03 mOhm	1	\$4.99	 WE-HCI_2212 588 mm ²
M1	NA	IdealFET	VdsMax= 320.0 V IdsMax= 5.0 Amps	1	NA	KCS0003B 80 mm ²
Rcomp	Vishay-Dale	CRCW0402374RFKED Series= CRCW..e3	Res= 374.0 Ohm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rfb1	Vishay-Dale	CRCW0402121RFKED Series= CRCW..e3	Res= 121.0 Ohm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rfb2	CUSTOM	CUSTOM Series= ?	Res= 21.0 kOhm Power= 0.0 W Tolerance= 0.0%	1	NA	CUSTOM 0 mm ²
Rs1	Vishay-Dale	CRCW0402100RFKED Series= CRCW..e3	Res= 100.0 Ohm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rs2	Vishay-Dale	CRCW04023K48FKED Series= CRCW..e3	Res= 3.48 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rsns	Stackpole Electronics Inc	CSR1206FK10L0 Series= ?	Res= 10.0 mOhm Power= 500.0 mW Tolerance= 1.0%	1	\$0.11	 1206 11 mm ²
Rt	Yageo	RC0201FR-0786K6L Series= ?	Res= 86.6 kOhm Power= 50.0 mW Tolerance= 1.0%	1	\$0.01	 0201 2 mm ²
Ruv1	Vishay-Dale	CRCW04021K27FKED Series= CRCW..e3	Res= 1.27 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Ruv2	Yageo	RC0201FR-0710KL Series= ?	Res= 10.0 kOhm Power= 50.0 mW Tolerance= 1.0%	1	\$0.01	 0201 2 mm ²
U1	Texas Instruments	LM5022QDGSTQ1	Switcher	1	\$2.59	 MUB10A 24 mm ²







Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	278.464 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	13.182 mW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	726.95 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	528.46 μ W	Capacitor	Output capacitor power dissipation
5.	IC Tolerance	25.0 mV	IC	IC Feedback Tolerance
6.	ICThetaJA	200.0 degC/W	IC	IC junction-to-ambient thermal resistance
7.	Iin Avg	3.622 A	IC	Average input current
8.	L Ipp	964.63 mA	Inductor	Peak-to-peak inductor ripple current
9.	L Pd	364.75 mW	Inductor	Inductor power dissipation
10.	L1 Irms	3.181 A	Inductor	Inductor ripple current
11.	Cin Pd	13.182 mW	Power	Input capacitor power dissipation
12.	Cout Pd	528.46 μ W	Power	Output capacitor power dissipation
13.	L Pd	364.75 mW	Power	Inductor power dissipation
14.	Rfb Pd	2.292 W	Power	Rfb Power Dissipation
15.	Rfb Pd	2.292 W	Resistor	Rfb Power Dissipation
16.	BOM Count	22	System	Total Design BOM count
17.	Cross Freq	334.969 Hz	System	Bode plot crossover frequency
18.	Duty Cycle	94.476 %	System	Duty cycle
19.	FootPrint	1.068 k mm ²	System	Total Foot Print Area of BOM components
20.	Frequency	200.055 kHz	System	Switching frequency
21.	Gain Marg	-25.025 dB	System	Bode Plot Gain Margin
22.	Iout	200.0 mA	System	Iout operating point
23.	Low Freq Gain	66.283 dB	System	Gain at 1Hz
24.	Mode	CCM	System	Conduction Mode
25.	Phase Marg	78.804 deg	System	Bode Plot Phase Margin
26.	Pout	44.0 W	System	Total output power
27.	SW Ipk	3.652 A	System	Peak switch current
28.	Total BOM	NA	System	Total BOM Cost
29.	Vin	14.0 V	System	Vin operating point
30.	Vout	220.0 V	System	Operational Output Voltage
31.	Vout Actual	219.938 V	System	Vout Actual calculated based on selected voltage divider resistors
32.	Vout Tolerance	3.008 %	System	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
33.	Vout p-p	23.234 mV	System	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	200.0 m	Maximum Output Current
VinMax	17.0	Maximum input voltage
VinMin	14.0	Minimum input voltage
Vout	220.0	Output Voltage
base_pn	LM5022-Q1	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

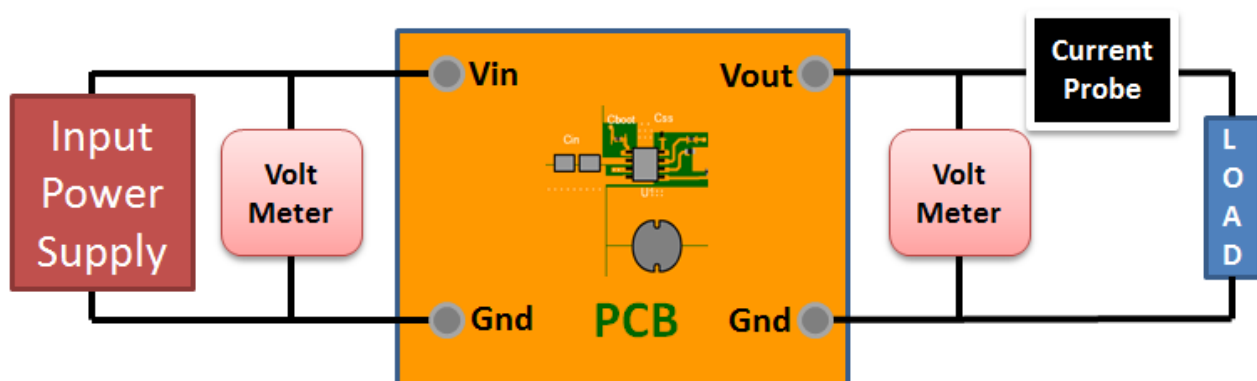
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 14.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum lout of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Feature Highlights: Automotive Qualified Synchronous Boost Controller
2. The LM5022-Q1 is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application
3. Master key : 97368DBC099F4C2A1B0A3C0B7CEDD87E[v1]
4. **LM5022-Q1** Product Folder : <http://www.ti.com/product/LM5022%2DQ1> : contains the data sheet and other resources.

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